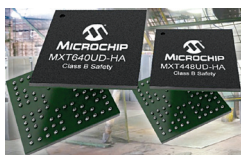


COMPONENTS

Capacitive touchscreen controller

Microchip MXT448UD-HA and MXT640UD-HA capacitive touchscreen controllers provide dedicated solutions for larger screen sizes as well as scalability. In addition to Class B, these new



devices are compliant with the IEC61000-4-6 Class A specification for conducted

noise immunity up to 10Vrms (level three for industrial use), enabling products with a touch screen interface to function in very harsh factory environments. Users of consumer electronics devices such as ovens or washing machines having unprotected touchscreens are now safe from false or missed touches caused by electromagnetic interference from unshielded power transmission, electric motors, and neighbouring electrical equipment.

MicrochipTechnology
www.microchip.com

3D Hall-effect position sensor

Texas Instruments has introduced an accurate 3D Hall-effect position sensor, the TMAG5170, enabling engineers to achieve uncalibrated ultra-high precision at speeds up to 20kSPS for faster and more accurate real-time control



in factory automation and motor-drive applications. The sensor also provides

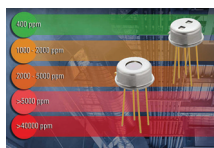
integrated functions and diagnostics to maximise design flexibility and system safety, while using at least 70% less power than comparable devices. Additionally, it increases safety with a unique set of smart diagnostic capabilities

such as checks for communication, continuity, and internal signal path as well as configurable diagnostics for the external power supply, magnetic field, and system temperature.

Texas Instruments
www.ti.com

Detectors for CO₂ sensors

Renesas has expanded its environmental sensor portfolio with the addition of a thermopile based detector series that has integrated optical filters for



optical (NDIR) CO₂ sensors. The four new single- and dual-channel analogue detectors for CO₂ gases

deliver good performance, quality, and longevity in a TO-5 package, making them ideal for use in a variety of industrial, medical, and IoT applications that require high reliability, high accuracy, and operation in high temperatures. The new detectors deliver fast response times ranging from 18 to 25ms and a high signal-to-noise ratio ranging from 1,981 to 4,953. The devices are certified for high temperature environments, supporting applications operating with extended temperature ranges up to 125°C.

Renesas Electronics Corporation
www.renesas.com

High image quality sensors

onsemi has released a new 8.3MP CMOS digital image sensor with rolling shutter and embedded high dynamic range (eHDR) technology. The



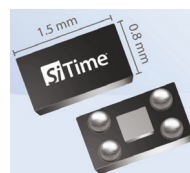
AR0821CS includes a 2.1µm DR-Pix backside illuminated (BSI) pixel design, which enables high quantum efficiency as well

as on-chip HDR (eHDR) technology for best-in-class dynamic range in excess of 140dB. The high-quality image data minimises processing needs, resulting in a faster and more cost-effective camera system. It meets the diverse needs of commercial, consumer, and industrial applications by providing superior image quality in challenging lighting conditions.

onsemi
www.onsemi.com

Wireless charging MEMS oscillator

SiTime has introduced the SiT3901 µPower digitally controlled MEMS oscillator (DCXO) targeting power-sensitive and space-constrained mobile and IoT



applications. The SiT3901 improves wireless charging speed by up to 25% while reducing the overall timing solution area by up to

90%. It is ideal for wireless charging systems for smartwatches, activity trackers, hearing aids, and wearables. Other features include programmable frequency of 1MHz to 26MHz, stability over temperature of ±50 and ±100 ppm, wide temperature range from -40°C to +85°C, and more.

SiTime Corporation
www.sitime.com

Clinical single-chip front-end

Medical device designers are eager to reduce annual healthcare-related expenditures by replacing office based health monitoring systems with smaller, lower



power, wireless devices worn discreetly and continuously in the home

or office. With three clinical-grade subsystems integrated into one IC, the single-chip MAX86178 triple system analogue front-end (AFE) from Analog